

WELDING AND FABRICATION

A welding and Fabrication course combines classroom theory with hands-on training in metal joining, cutting, and shaping, covering key techniques like SMAW (stick), GMAW (MIG), and GTAW (TIG). Programs typically include blue print reading, safety protocols, material science, and structural fabrication, preparing students for roles in manufacturing, construction, and repair.

Core Curriculum Components

- **Welding Processes:** In-depth training on Manual Metal Arc Welding (MMAW/Stick), Metal Active Gas (MAG/MIG), Tungsten Inert Gas (TIG), and Oxy-acetylene welding.
- **Fabrication Techniques:** Techniques for cutting, bending, forming, and assembling metal components, including structural steelwork and sheet metal work.
- **Technical Drawing and Blueprints:** Reading and interpreting engineering drawings, welding symbols, and specifications.
- **Workshop Technology & Safety:** Training on machinery maintenance, proper use of Personal Protective Equipment (PPE), and safety regulations.
- **Material Science:** Understanding the properties of metals (aluminum, stainless steel, mild steel) and their reaction to heat.
- **Quality Control & Inspection:** Identifying weld defects (visual inspection) and ensuring structural integrity.

Module Breakdown

1. **Introduction to welding & Safety:** Fundamentals, safety regulations, and equipment maintenance.
2. **General metalwork & Fabrication:** Measuring, marking, cutting, bending, and shaping metal.
3. **Advanced Welding Techniques:** Specialized, high precision welding on diverse materials.
4. **Structural Fabrication & Pipe Welding:** Project –based training on creating frameworks, tanks, or structural components.
5. **Engineering Mathematics & science:** Applied calculations for welding projects.

Certification and Diploma Levels

- **Certificate/NITA Level:** Focuses on foundational practical skills, often 6-12 months
- **Diploma Level:** Includes advanced welding, computer – aided design (CAD), and industrial management, typically lasting 2-3 years

Important Note: are you looking for a certificate program, a diploma program, or short NITA-certified courses? Knowing if you are interested in a specific, shorter course or a

longer, more in-depth program will help me narrow down the options.